PROJECT REPORT
VoltRelay Data Analysis

Exploratory data analysis, statistical modelling, and telemetry trend visualisation for a battery-swapping energy network using Jupyter notebooks and data-science libraries.

Project Title	VoltRelay Energy — Battery Swapping Network Analysis
Student / Owner	Tejas Gowda M
Repository	github.com/tejas448866/VoltRelay-Data-Analysis
Deliverable	Jupyter notebook (VoltRelay_Data_Analysis.ipynb)

1. Project Purpose

VoltRelay-Data-Analysis is an end-to-end data analytics project for a battery-swapping energy network (VoltRelay Energy). The notebook examines operational telemetry and business tables to surface actionable insights on network health, service quality, equipment performance, pricing, and rider behaviour.

The primary objective is to analyse network performance, service failures, station and geographic patterns, battery and equipment reliability, pricing and partner economics, and rider retention / churn. Findings are intended to support operational decisions such as station placement, inventory planning, supplier evaluation, and retention campaigns.

Target users are data analysts, operations managers, and students learning applied exploratory data analysis (EDA) on multi-table operational datasets. The work is delivered as a reproducible Jupyter notebook that can be run locally or on Google Colab.

2. Technologies Used

The following technologies are used in the notebook and analysis workflow:

- **Python 3** — primary language for data processing and visualisation.
- **pandas** — loading, cleaning, joining, and aggregating tabular datasets.
- **NumPy** — numerical operations and handling of invalid sensor values.
- **matplotlib** — charts for trends, distributions, and bar comparisons.
- **seaborn** — statistical visualisations and cleaner default styling.
- **Jupyter Notebook** — interactive, documented analysis environment (Colab-compatible).
- **CSV operational data** — batteries, stations, riders, swap events, tickets, and context tables.

No machine-learning model training is required for the core notebook; the emphasis is on rigorous cleaning, validation, aggregation, and clear visual reporting.

3. Working and Logic

Data Sources

The analysis loads multiple related tables that together describe the swapping network:

- **swap_events** — individual battery swap attempts (success / failure, SOC, SOH, km, firmware, sync mode).
- **stations** / **station_hourly_status** — station locations and hourly availability telemetry.
- **batteries** — pack type, supplier, manufacturing lot, and current state of health (SOH).
- **riders** — rider profiles, signup dates, and home cities.
- **support_tickets** — customer service interactions linked to failures and experience.
- **fleet_partners** — partner economics and fleet relationships.

- **city_daily_context** — city-level daily context for demand and environment.

Analysis Workflow

1. Import libraries and load all CSV tables into pandas DataFrames.
2. Profile shapes, missing values, and basic distributions for every table.
3. Clean swap events: clamp or null invalid km, SOC, and SOH readings; normalise city names.
4. Convert timestamp columns to datetime and validate date ranges (Jan–May 2024 in the sample).
5. Investigate data-quality issues (firmware timestamp anomalies, offline-batch near-duplicates).
6. Aggregate network performance over time (volume, success rate, hourly patterns).
7. Analyse service failures (e.g. failed_no_charged_battery) by hour, station, and geography.
8. Evaluate battery and equipment performance by supplier, manufacturing lot, and SOH distribution.
9. Explore pricing / partner economics and rider retention indicators where data supports it.
10. Produce charts and summary tables that support business recommendations.

Core Analysis Areas

- **Network performance over time** — volume trends, completion rates, temporal seasonality.
- **Service failures & customer experience** — failure types, support tickets, peak failure hours.
- **Station & geographic patterns** — city normalisation, station utilisation, regional differences.
- **Battery & equipment performance** — SOH distribution, supplier failure rates, lot-level quality.
- **Pricing & partner economics** — revenue and cost signals where present in the tables.
- **Rider retention & churn** — activity patterns and retention proxies from rider and swap history.

Decision Logic & Data Quality

Invalid sensor values (negative kilometres, SOC/SOH above 100%) are set to missing rather than deleted, preserving event counts while protecting averages. City strings are mapped to canonical names (e.g. HYD / Hyderabad → Hyderabad). Firmware v3.2.0 timestamp concerns documented for a later period are checked against the sample range and skipped when not applicable. Offline-batch sync rows are scanned for exact and near-duplicate keys before aggregation so that double-counted events do not inflate metrics.

4. Testing and Limitations

Validation is performed inside the notebook through shape checks, missing-value reports, value-range inspections, and targeted filters (firmware windows, duplicate detection). Charts and group-by tables act as visual and tabular tests of the cleaning steps. The notebook is Colab-ready via a badge link for independent re-execution.

Challenges Observed

- Inconsistent city naming across rider and station tables requires explicit mapping.
- Sensor outliers (SOC, SOH, km) appear in production-like telemetry and must be handled carefully.
- Offline batch uploads can introduce near-duplicates that bias volume and failure rates.
- Sample date coverage may not include every documented firmware incident window.

Limitations

- Analysis quality depends on the completeness and accuracy of the provided CSV extracts.
- Causal claims (e.g. supplier quality) need operational confirmation beyond observational rates.
- Rider churn definitions are approximate when explicit churn labels are absent.
- The repository currently ships the notebook and a minimal README; raw CSVs are assumed external or Colab-uploaded.

Possible Future Improvements

- Package cleaning steps into reusable functions or a small ETL module.
- Add predictive models for failure risk and rider churn on top of the EDA.
- Automate dashboard exports (e.g. Streamlit or static HTML) for operations teams.

- Include sample anonymised CSVs or a data dictionary in the repository for easier onboarding.

5. GitHub and Originality

Repository Structure

- **README.md** — project title and entry point.
- **VoltRelay_Data_Analysis.ipynb** — full analysis notebook (loading, cleaning, EDA, charts).

The notebook is self-contained for analysis once the CSV datasets are available in the runtime environment (local path or Colab /content).

Execution Steps

1. Clone the repository: `git clone https://github.com/tejas448866/VoltRelay-Data-Analysis`
2. Install Python 3 and the stack: `pip install pandas numpy matplotlib seaborn jupyter`
3. Place the operational CSV files where the notebook expects them (or adjust paths).
4. Open `VoltRelay_Data_Analysis.ipynb` in Jupyter or Google Colab.
5. Run cells top-to-bottom; review intermediate tables and charts for each analysis area.
6. Export key figures or summary tables as needed for reports or stakeholders.

Individual Contributions

I, Tejas Gowda M, contributed to the overall project development, including data loading and cleaning logic, exploratory analysis across the six core areas, visualisation, validation of data-quality issues, and documentation for academic submission.

AI Assistance

AI tools were used for coding assistance, architecture understanding, and preparing documentation during development and report writing.

Learning Outcomes

- Multi-table EDA on operational energy / mobility telemetry.
- Practical data-quality handling (outliers, duplicates, inconsistent categories, timestamps).
- Aggregating and visualising failure rates by time, geography, supplier, and lot.
- Structuring a notebook so that cleaning, analysis, and insights remain reproducible.
- Communicating business-oriented findings from technical sensor and event data.

Conclusion

VoltRelay-Data-Analysis successfully demonstrates a complete analytics pipeline for a battery-swapping network: from multi-source data loading and careful cleaning, through network, failure, station, battery, pricing, and retention analysis, to clear visual summaries. The Jupyter-based approach keeps the work transparent and reproducible, making it both a practical decision-support artefact and an educational example of applied data science for energy and mobility operations.

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